Topic: Functions

Part 1 Introduction to Functions

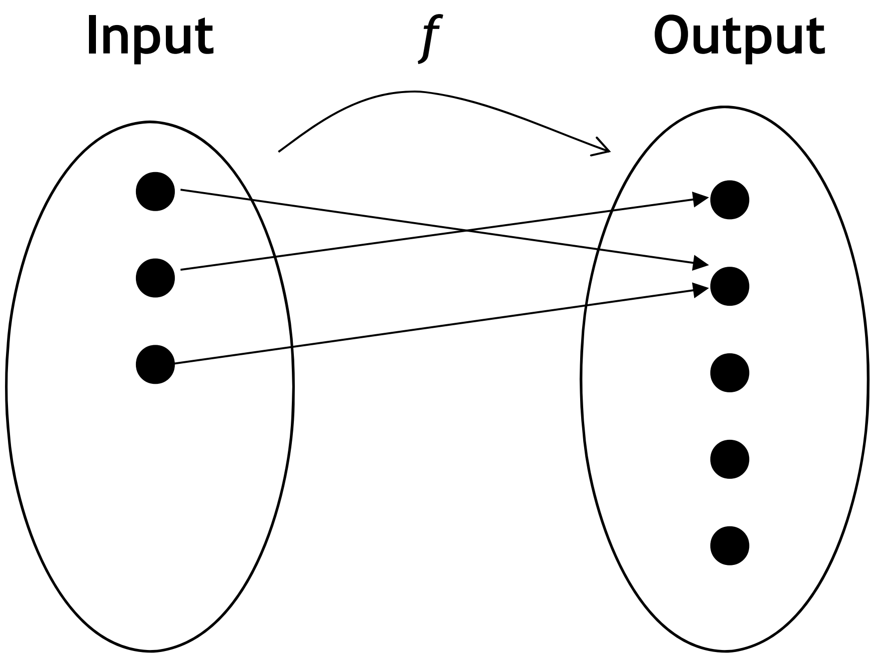
Section 1
Introduction

- Introduction
- Functions Defines by Formulas
- Functions Related To Algorithm Efficiencies

Introduction to Functions

Suppose two sets of objects are given – a first set and a second set – and suppose that within each element of the first set is associated with a particular element of the second set.

The relationship between the elements of the sets is called a **function.**



Arrow Diagram

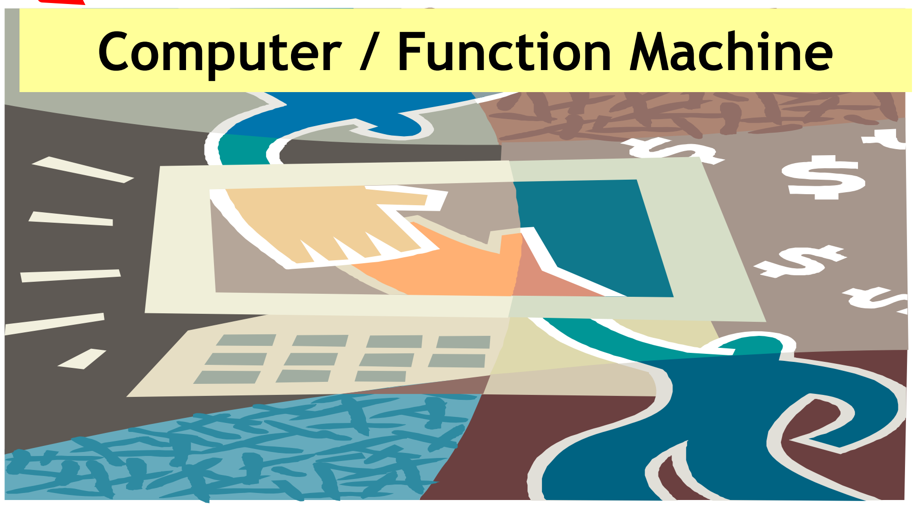
Function Machines

Another useful way is to think of a function as a machine (or computer).

Suppose f is a function from X to Y and an input X is given. Imagine f producing the output f(x) after processing x through a certain function.

Program Input

x

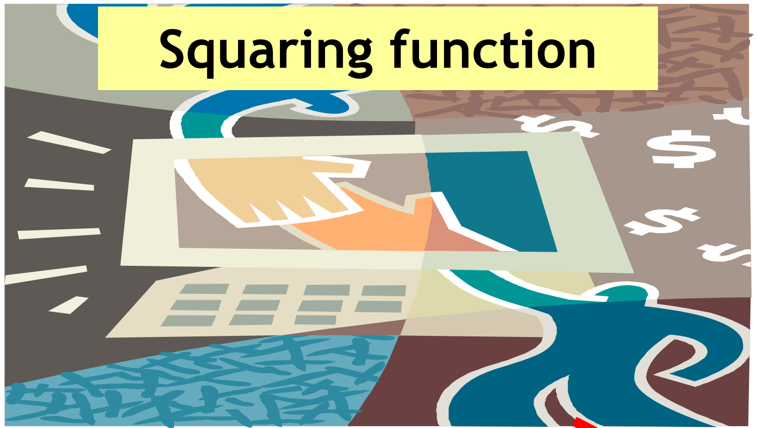


Program Output

f(x)

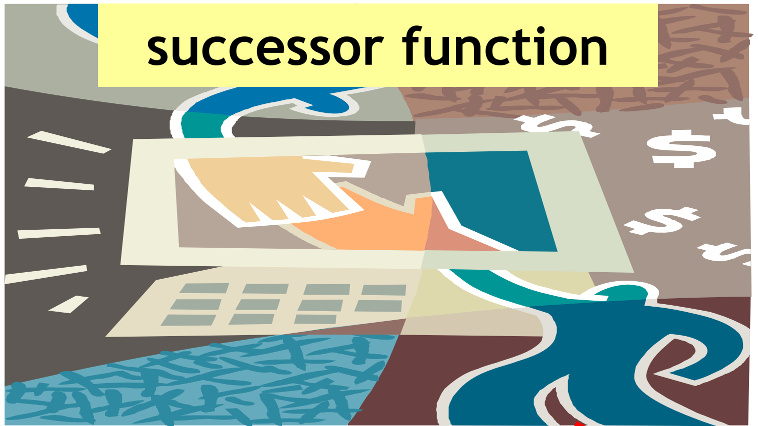
Functions Defined by Formulas

x



$f(x) = x^2$

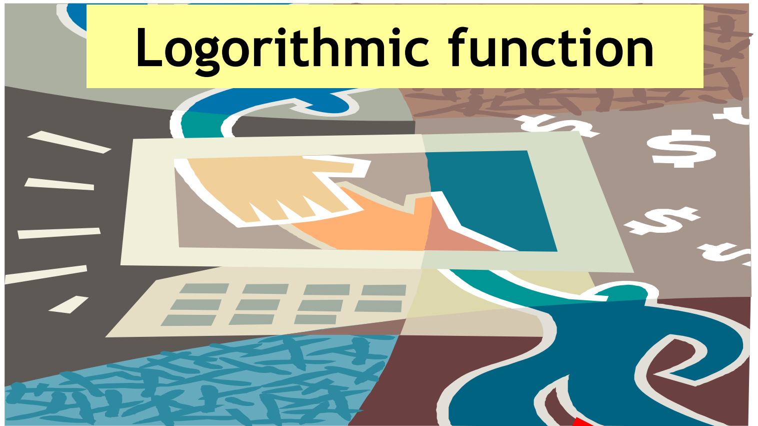
k



$g(k) = k+1$

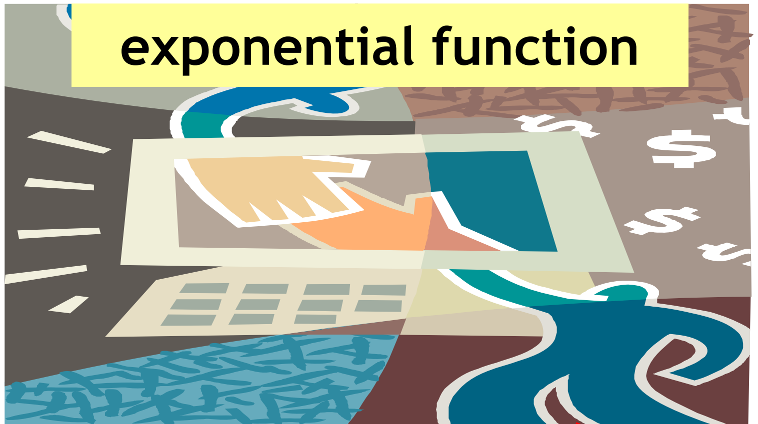
Functions Related To Algorithm Efficiencies

x

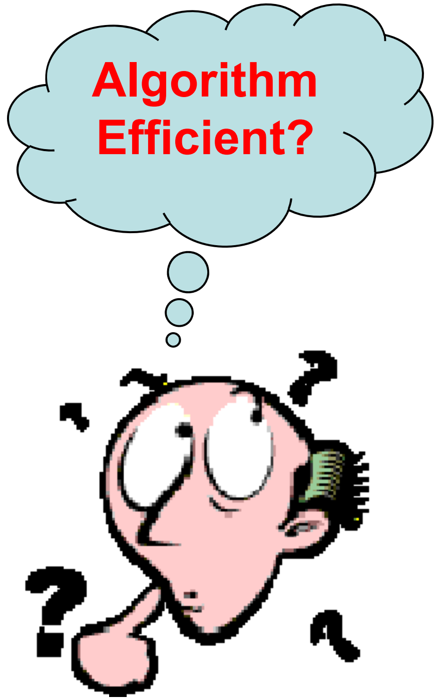


$f(x) = \log x$

k



$g(k) = e^k$



Section 2
Terminologies Relating To Functions

- Introduction
- Notation
- Domain, Co-Domain, Range
- Image and Inverse Image
- Ordered Pairs

Terminologies Relating To Functions

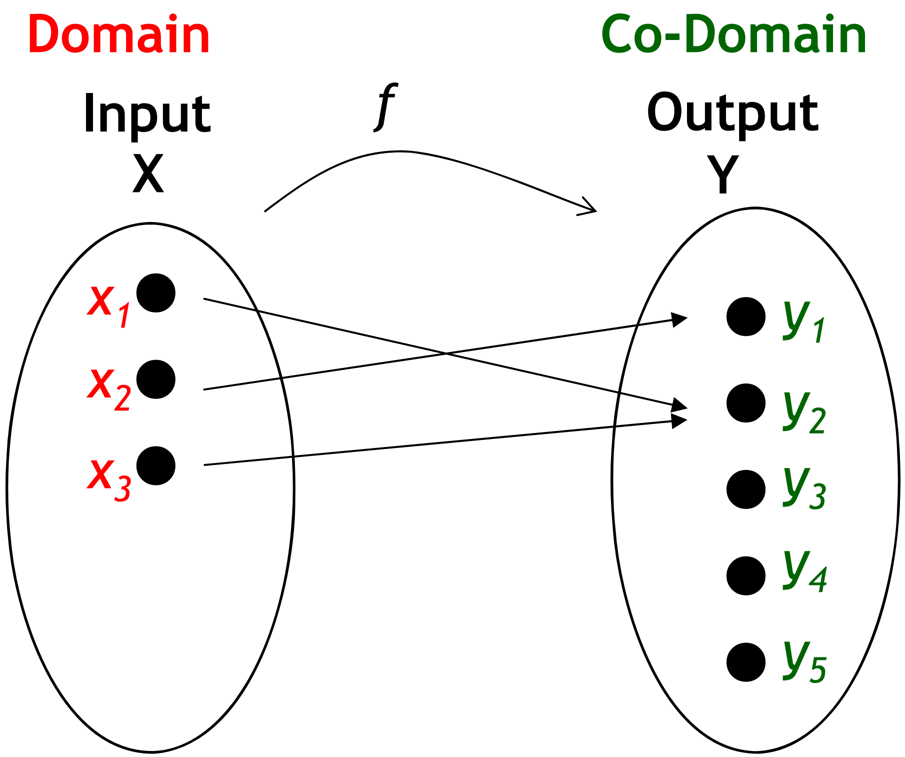
Let's start off with some terminologies relating to functions :

- Notation of $f: X \rightarrow Y$
- Domain
- Co-Domain
- Range
- Image
- Pre-image or Inverse Image
- Ordered Pairs

Notation of f : X → Y, Domain, Co-Domain

$$f : X \rightarrow Y$$

A **function *f* from a set X to a set Y** displays a relation between elements of X, called **INPUTS**, and elements of Y called **OUTPUTS**, with **each input is related to <u>one and only one output</u>.**



Domain of f = X

= { x_1, x_2, x_3 }

Co-domain of f = Y

= { y_1, y_2, y_3, y_4, y_5 }

Image of x and Range of f

Given an input element $x \in X$, if there is a unique output element $y \in Y$ related to x by *f*, then y is the **image of x under** *f*, denoted by ***f(x)*.**

The set of all values of *f* is called the **RANGE**.
Range of f = $\{y \in Y \mid y = f(x), \text{ for some } x \text{ in } X\}$

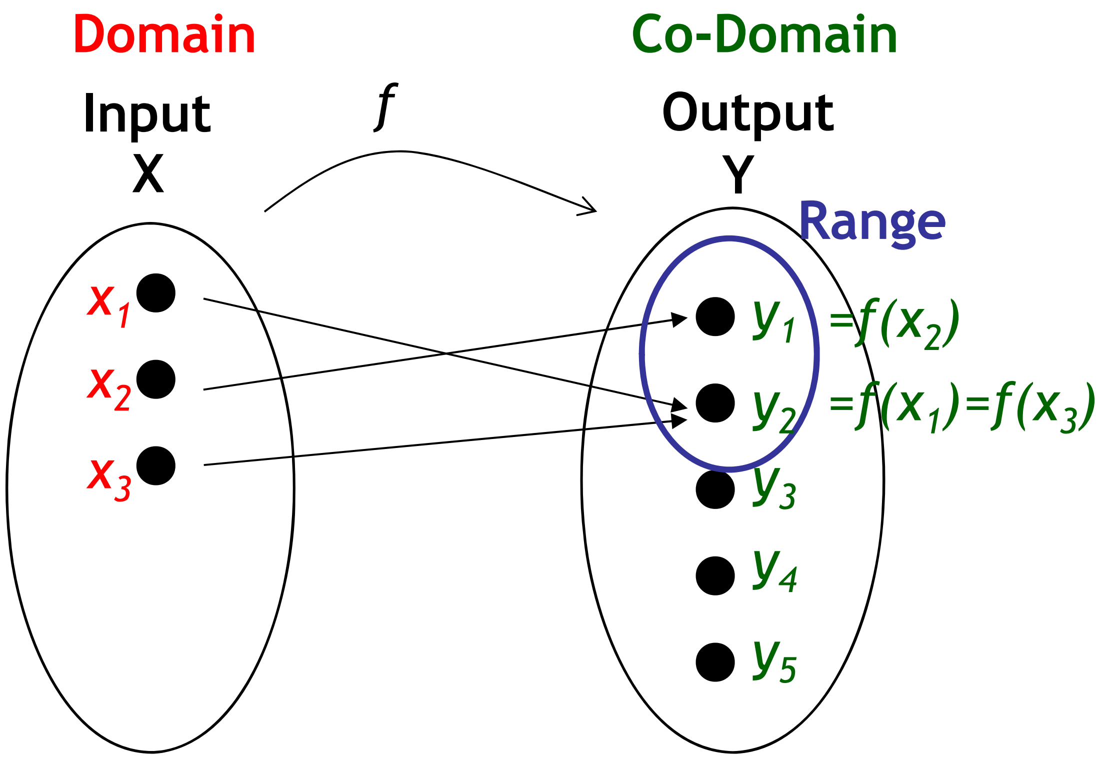


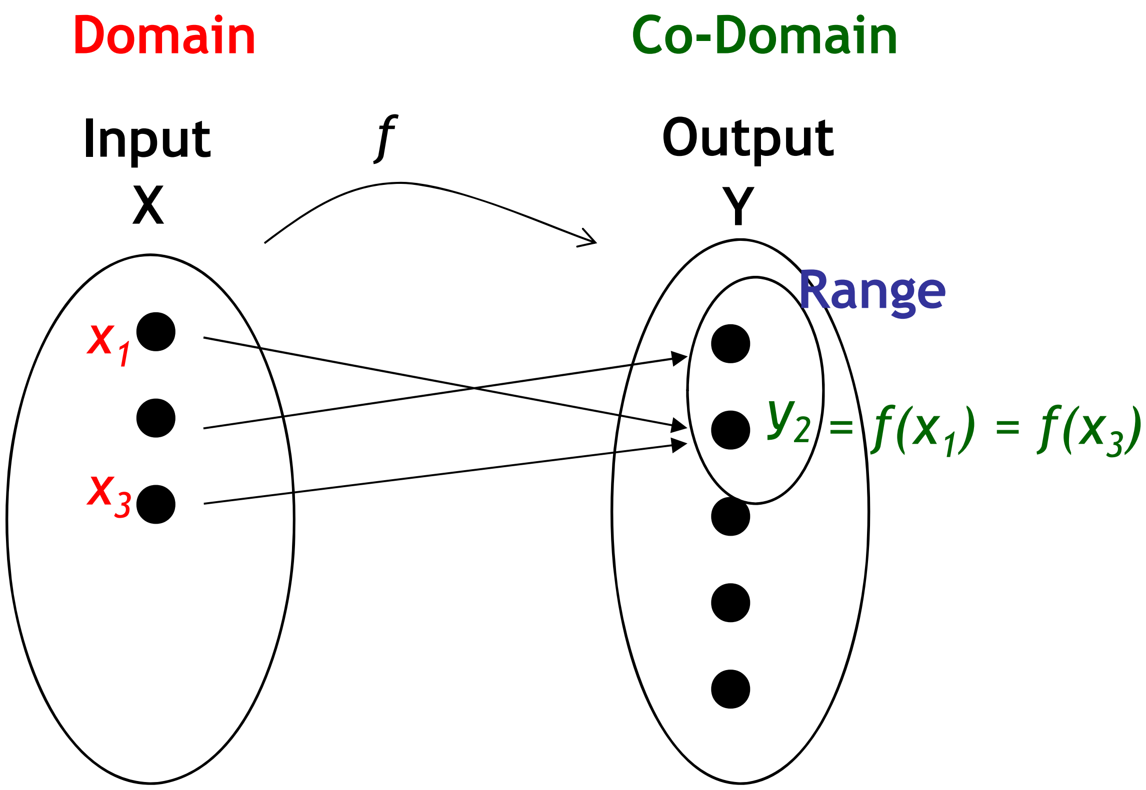
Image of x_1 = y_2 = $f(x_1)$
Image of x_2 = y_1 = $f(x_2)$
Image of x_3 = y_2 = $f(x_3)$

Range of f = $\{ y_1, y_2 \}$

Preimage or Inverse Image

Given an element y in Y,
there may exist elements in X such that f(x) = y,
then x is called a **PREIMAGE of y** or an **INVERSE image of y**.

Inverse Image of y = $\{x \in X \mid f(x) = y\}$



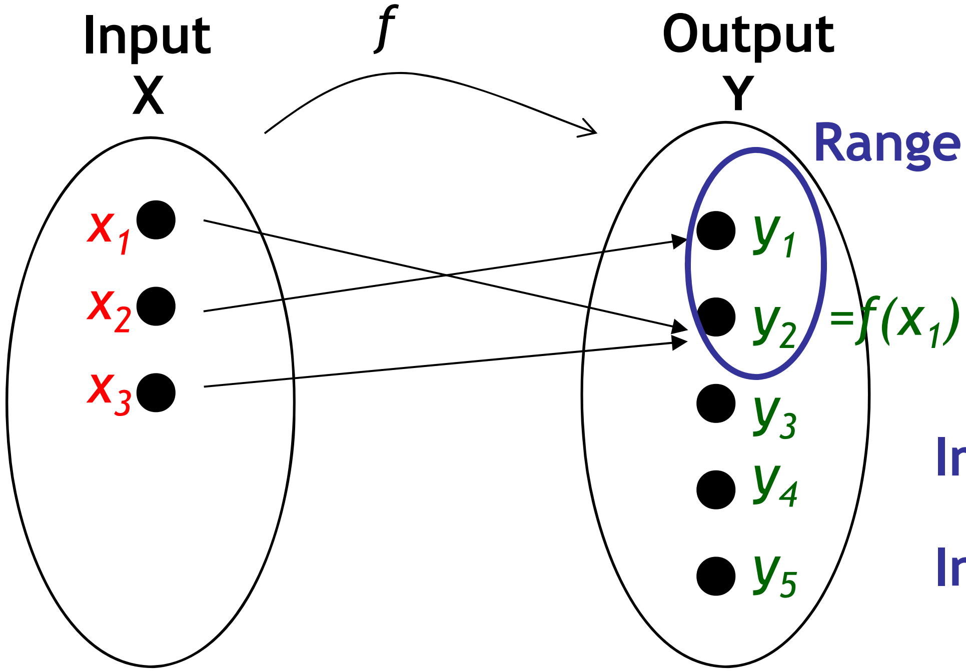
Inverse Image of y_2
$= \{ x_1, x_3 \}$

Summary of Terminologies

$f : X \to Y$ is a function that relates set X to set Y

Each input in X is related to <u>one and only one output in Y</u>.

Domain $= \{x_1, x_2, x_3\}$ **Co-Domain** $= \{y_1, y_2, y_3, y_4, y_5\}$



Range $= \{ y_1, y_2 \}$

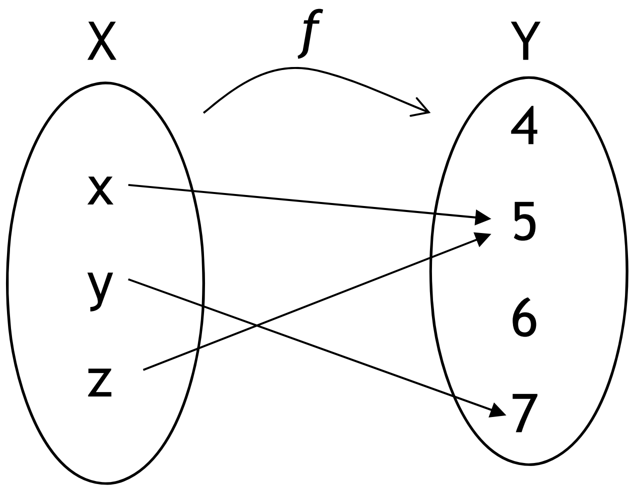
Image of $x_1 = y_2 = f(x_1)$

Inverse Image of $y_2 = \{x_1, x_3\}$

Ordered Pair : $\{ (x_1, y_2), (x_2, y_1), (x_3, y_2) \}$

Example (Terminologies)

Let X = {x, y, z} and Y = {4, 5, 6, 7}.
Define f: X→Y by the following arrow diagram.



(a) List the domain and co-domain of f.
(b) Find f(x), f(y) and f(z).
(c) What is the range of f?
(d) Find the inverse images of 5, 7 and 4.
(e) Represent f as a set of ordered pairs.

(a) Domain of f = {x, y, z}
Co-domain of f = {4, 5, 6, 7}

(b) f(x) = 5
f(y) = 7
f(z) = 5

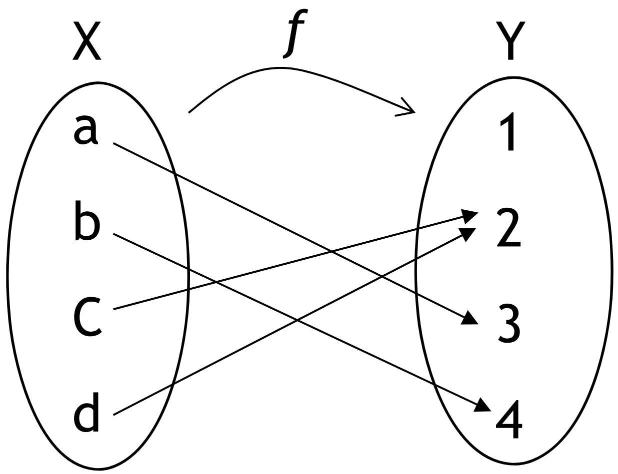
(c) Range of f = {5, 7}

(d) Inverse image of 5 = {x , z}
Inverse image of 7 = {y}
Inverse image of 4 = {}

(e) {(x, 5), (y, 7), (z, 5)}

Exercise (Terminologies)

Let X = {a, b, c} and Y = {1, 2, 3, 4}.
Define f: X→Y by the following arrow diagram.



(a) List the domain and co-domain of f.
(b) Find the image of a, b, c, d.
(c) What is the range of f?
(d) Find the inverse images of 1, 2, 3 and 4.
(e) Represent f as a set of ordered pairs.

(a) Domain of f = {a, b, c, d}
Co-domain of f = {1, 2, 3, 4}

(b) f(a) = 3
f(b) = 4
f(c) = 2
f(d) = 2

(c) Range of f = {2, 3, 4}

(d) Inverse image of 1 = { }
Inverse image of 2 = {c, d}
Inverse image of 3 = {a}
Inverse image of 4 = {b}

(e) {(a, 3), (b, 4), (c, 2), (d, 2)}

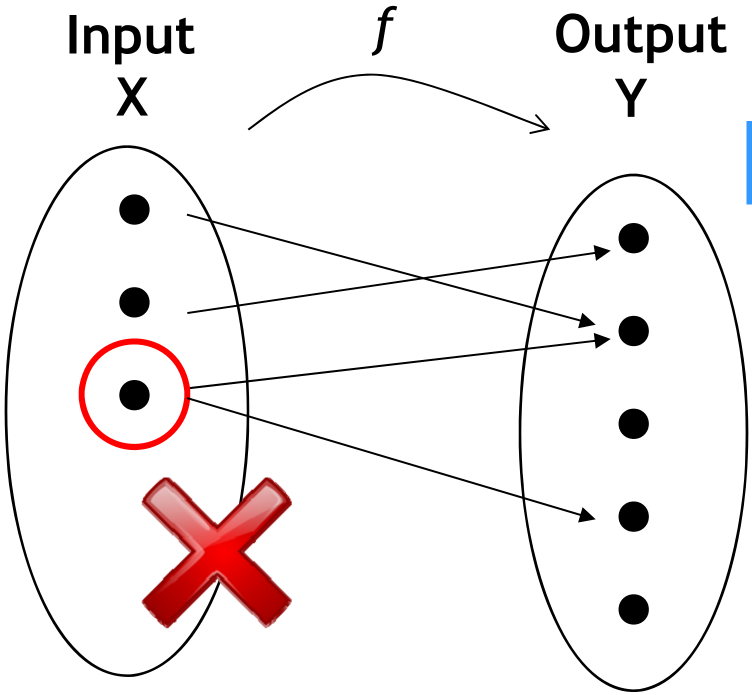
Part 1 Introduction to Functions

Section 3
Is It A Function?

- Introduction
- Examples & Exercises - Is It A Function?
- Example – Disprove by Showing 2 Images
- Example – Disprove by Showing No Image

Is it a Function?

A **function** *f* **from a set X to a set Y** displays a relation between elements of X, called **INPUTS**, and elements of Y called **OUTPUTS**, with **each input is related to <u>one and only one output</u>.**



Is it a function?

Every input has **exactly** 1 output.

How to Disprove?

There exist an input such that

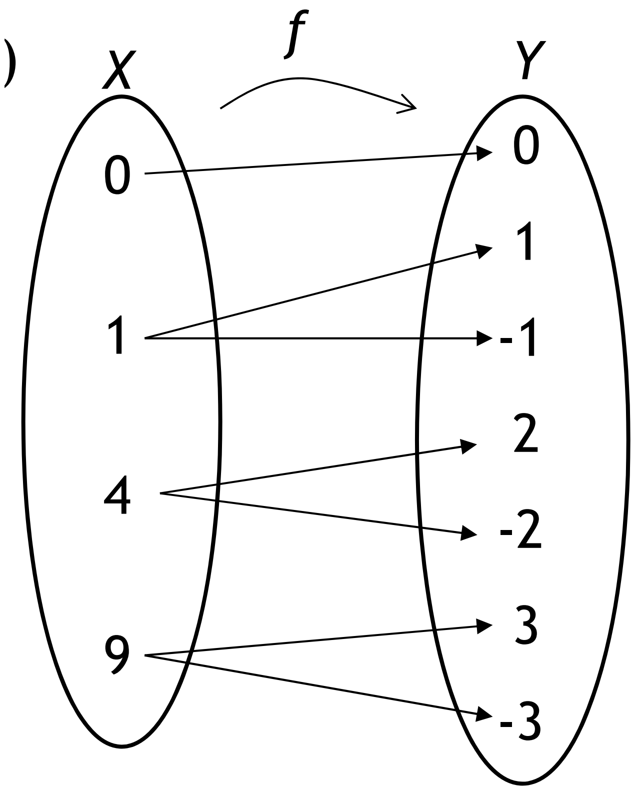
1. the input **does not** have an output

<u>OR</u> 2. the input has **more than 1** output

Example (Is it a Function)

Which of the following are functions? Justify with reason(s).

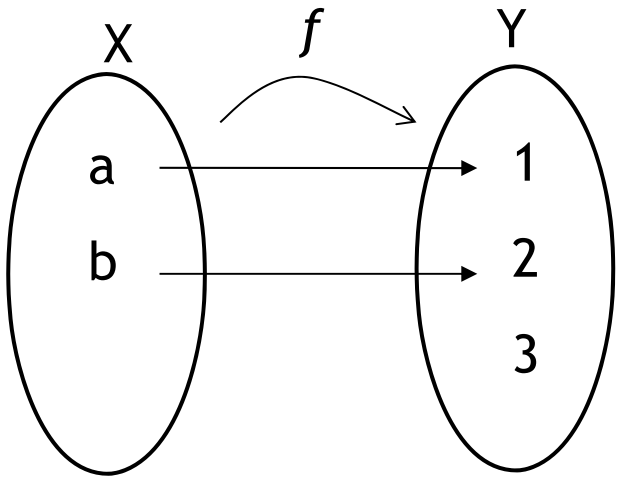
(a)



No, this is not a function.

Elements 1,4,9 in the domain has 2 images each.

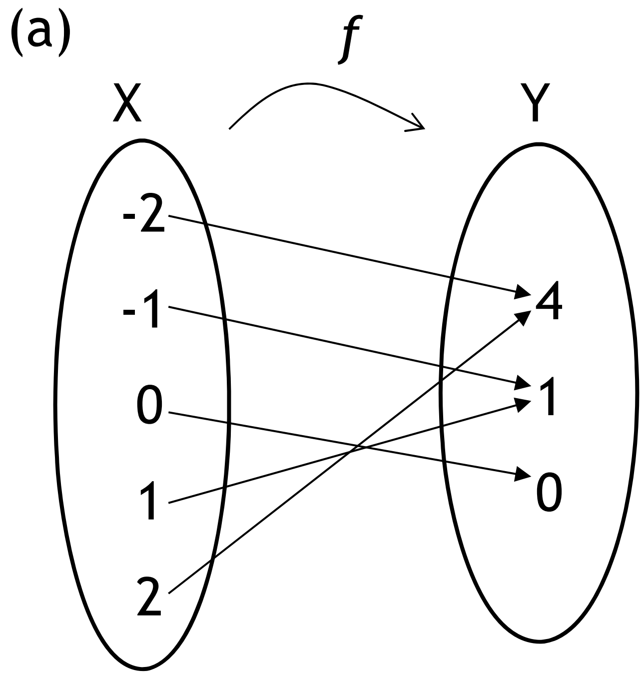
(b)



Yes, this is a function.

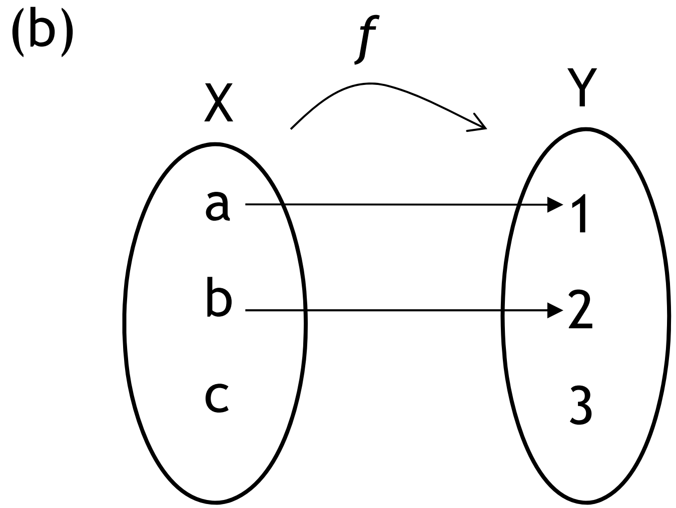
Every element in domain has only one image.

Exercise (Is it a Function)



Yes, this is a function.

Every element in domain has only one image.



No, this is not a function.

Not all elements in domain has an image.
"c" has no image.

Example (Disprove by Showing 2 Images)

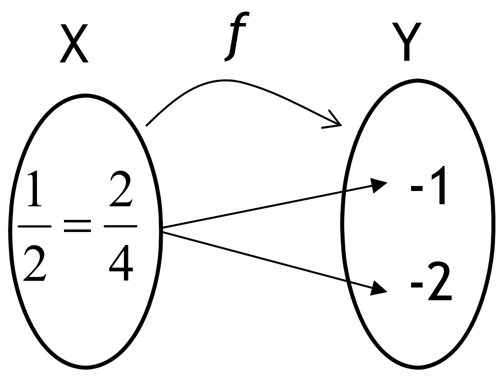
Q represents the set of all rational numbers.
Z represents the set of all integers.

A function f is defined as f: Q $\rightarrow$ Z such that $f\left(\frac{m}{n}\right) = m - n$
for all integers m and n with n≠0.

Is f a function?

Solution :
Fractions can have more than one representations: 0.5 = $\frac{1}{2}$ = $\frac{2}{4}$

But f($\frac{1}{2}$) = 1 – 2 = -1 while f($\frac{2}{4}$) = 2 – 4 = -2.
Thus, f($\frac{1}{2}$) ≠ f($\frac{2}{4}$)



This shows that the point 0.5 = $\frac{1}{2}$ = $\frac{2}{4}$ has 2 images.
Therefore, f is not a function.

Example (Disprove By Showing No Image)

R represents the set of real numbers

Define $f(x) = y$ if $x^2 + y^2 = 1$ where $X = Y = R$

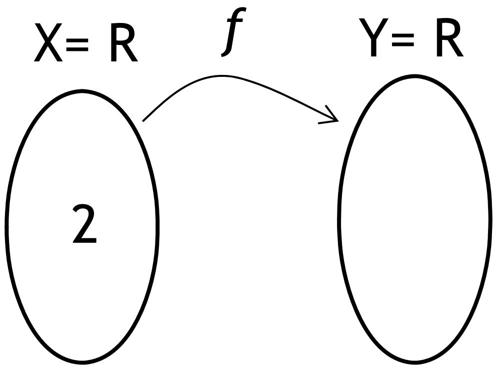
Is f a function?

Solution:

When x=2, $x^2 + y^2 = 1$

$$2^2 + y^2 = 1$$
$$y^2 = 1 - 2^2$$
$$y^2 = -3$$

Hence, there is no solution for y.



This shows that $2 \in X$ **does not have an image** in Y.
Therefore, f is **not a function.**

Summary (Is it a Function)

To prove that something is not a function, we need to show

- there exist $x \in X$ such that x **does not** have an image in Y
- or x has **more than one** image in Y.

Application : Function Defined on Strings

In computing, a string is a sequence of characters.

Let **S** be the set of all strings.
Let ϵ be the null string (the sting with no characters).

Define length : S → Z as g(s) = the number of characters in the string.

Find (a) length(cat) (b) length(hippopotamus) (c) length(**Є**)

Solution :

(a) 3 (b) 12 (c) 0

Application : Encoding and Decoding Functions

Digital messages are made up of 0's and 1's. When transmitted, they are usually coded in special ways to reduce interference noise, e.g. write each bit 3 times.

At the receiver's end, if one of the three consecutive bits is not identical, then error is detected. Otherwise, message is decoded by replacing each of the 3 repeated bits by 1 single bit.

Let E(s) = string obtained by writing each bit 3 times.

Let D(s) = string obtained by replacing each of the 3 repeated bits by one.

<u>Example</u>

Suppose the message to be transmitted is s = 011001.

The encoded message E(s) = 000111111000000111.

The decoded message D(s) = 011001.

Application : Boolean Functions

Digital logic circuits can be represented as input/output tables.

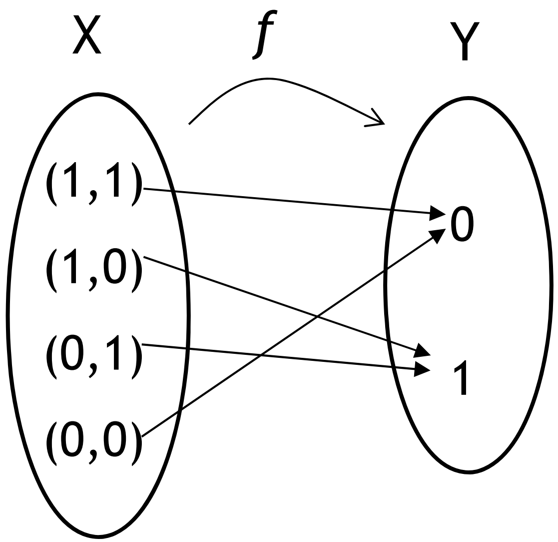
Such input/output tables are really just functions where :

Domain : Elements in the input column (ordered tuples of 0's and 1's).

Co-domain : Elements in the output column (either 0 or 1)

Relation : Function sends the each input element to the output element in the same row.

Input		Output
x_1	x_2	$f(x_1,x_2)$
1	1	0
1	0	1
0	1	1
0	0	0



Part 2 One-to-one, Onto & Inverse Functions

Section 1
Introduction

- Introduction to One-to-one and Onto Functions
- Introduction to Inverse Functions

Introduction

Two important properties that functions may satisfy:

- One- to-one
- Onto

When a function is both one-to-one and onto,

we can define an **inverse function**

that "**undoes**" the action of the function

from the co-domain to the domain.

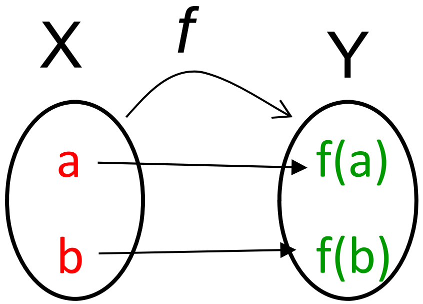
Section 2
One-to-one Functions

- When is a Function Considered One-to-one?
- Example and Exercise

When is a Function Considered One-to-One?

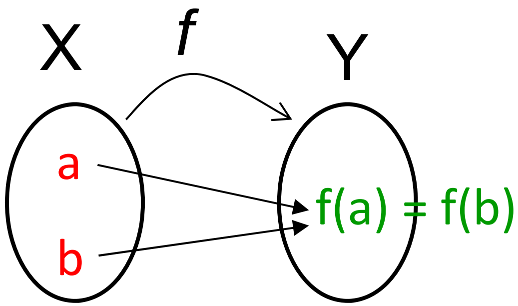
f is **one-to-one**

- if **every input** has **an unique output; or**
- if every element in the domain has a **distinct image.**



f is **not one-to-one**

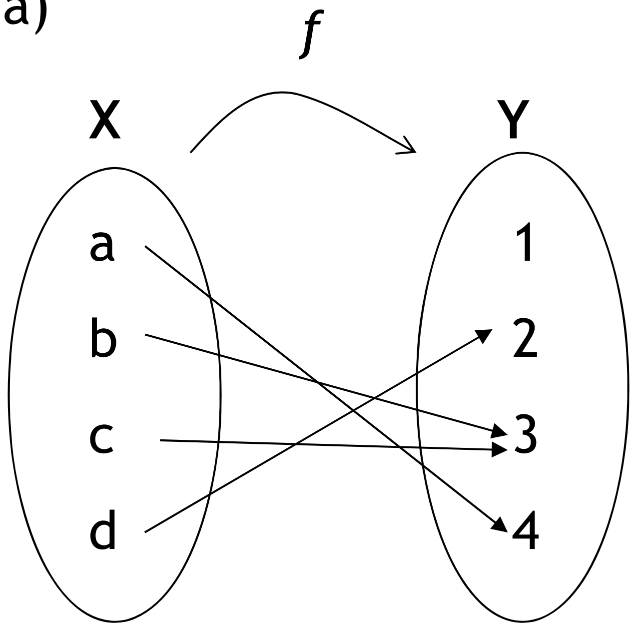
- if some elements in the domain **share** the **same image.**



Example (One-to-One Function)

Which of the following functions below are one-to-one?

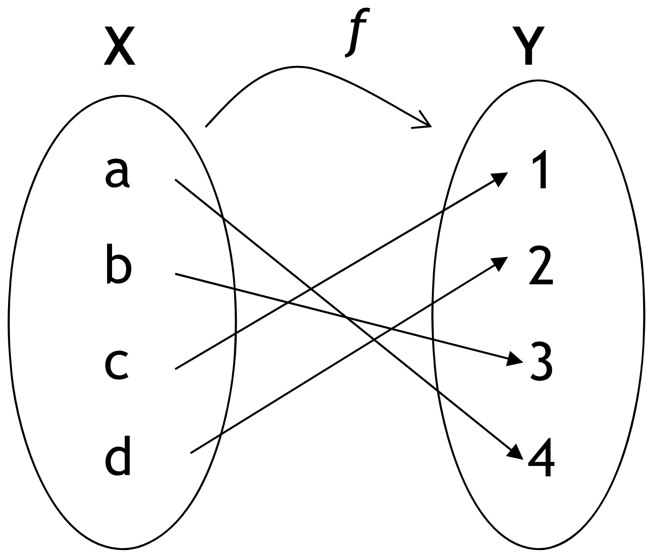
(a)



No, f is not one-to-one

“b” & “c” share the same image.

(b)



Yes, it is one-to-one

<u>All elements</u> in the <u>domain</u> has one <u>unique</u> image.

Exercise (One-To-One Function)

Let X = {1, 2, 3} and Y = {p, q, r, s}.

(a) Define f: X→Y as follows: $f(1) = r$, $f(2) = q$ and $f(3) = p$.
Is f one-to-one? Explain.

(b) Define g: X→Y as follows: $g(1) = s$, $g(2) = r$ and $g(3) = s$.
Is g one-to-one? Explain.

Solution :

(a) *f is one-to-one*
Reason : Every element in the domain has an unique image.

(b) g is not one-to-one
Reason: Elements 1 and 3 in the domain share the same image "s".

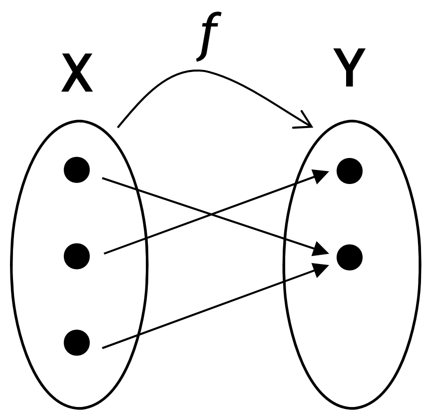
Section 3
Onto Functions

- When is a Function Considered Onto?
- Example and Exercise

When is a Function Considered Onto?

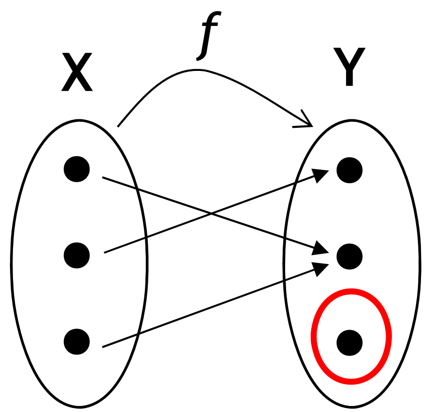
f is **onto**

- If every element in the co-domain has a pre-image.



f is **not onto**

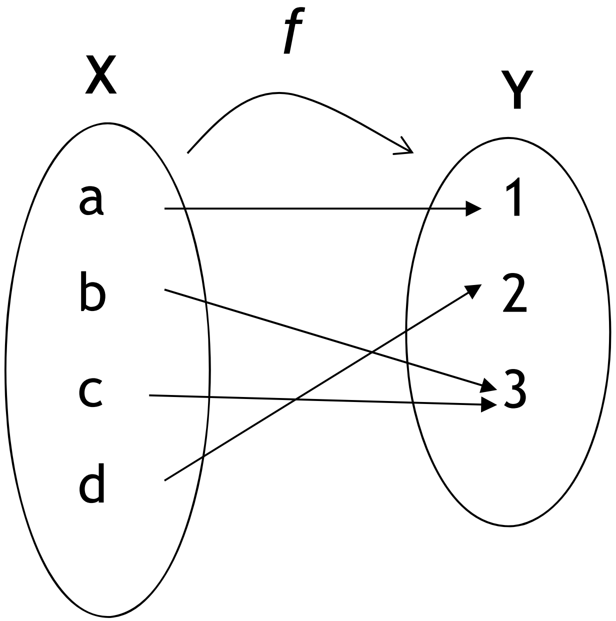
- If some elements in the co-domain do not have a pre-image.



Example (Onto Function)

Which of the following functions below are onto?

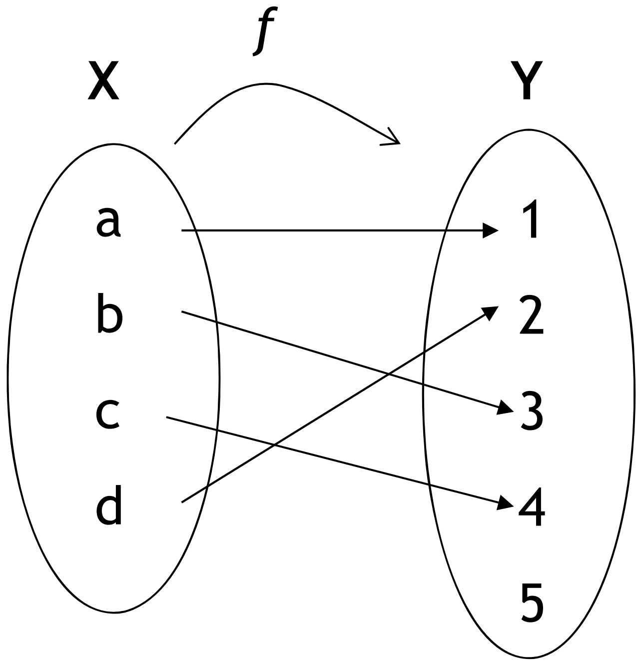
(a)



Yes, onto

Each element in the co-domain has a pre-image.

(b)



No, not onto

Element 5 in the co-domain does not have a pre-image.

Exercise (Onto Function)

Let X = {1, 2, 3, 4} and Y = {p, q, r}.

(a) Define f: X→Y as follows: $f(1) = p$, $f(2) = q$, $f(3) = p$ and $f(4) = r$.
Is f onto? Explain.

(b) Define g: X→Y as follows: $g(1) = q$, $g(2) = r$, $g(3) = r$ and $g(4) = q$.
Is g onto? Explain.

Solution :

(a) *f is onto*
Reason : Every element in the co-domain has a pre-image.

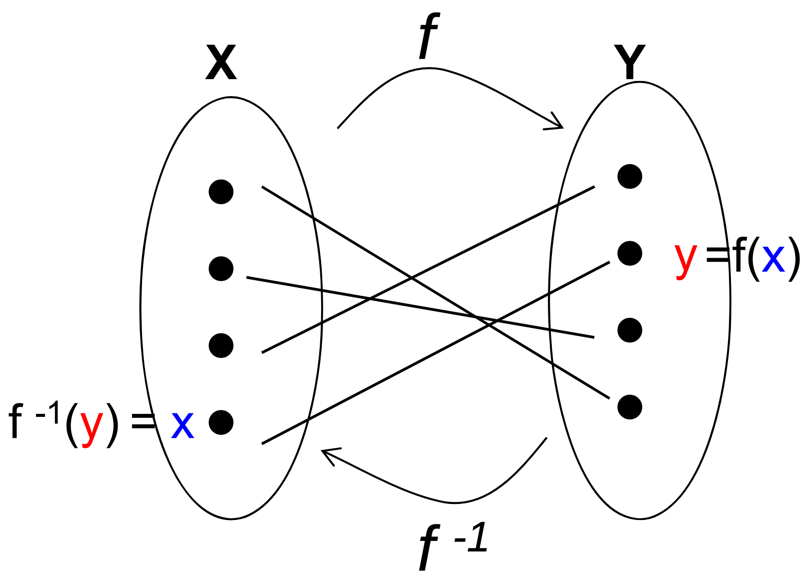
(b) g is not onto
Reason: Element p in the co-domain does not have a pre-image.

Section 4
Inverse Functions

Introduction to Inverse Function

If f: X → Y is both one-to-one <u>and</u> onto,
then there exists a function f^{-1} : Y → X that "undoes" the action of f
(i.e. f^{-1} sends elements in Y back to elements in X).

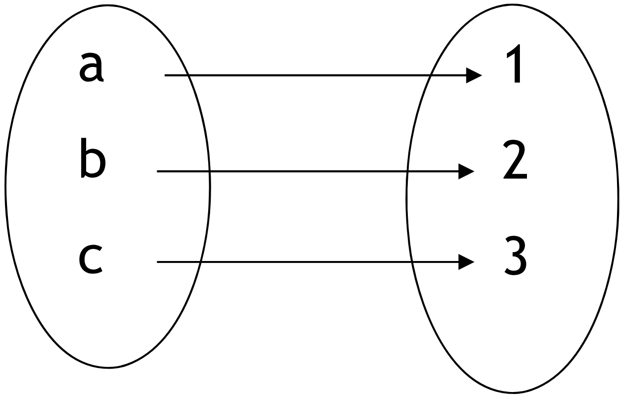
This function f^{-1} is called the <u>inverse</u> function for f.



Example (Arrow Diagram of Inverse Function)

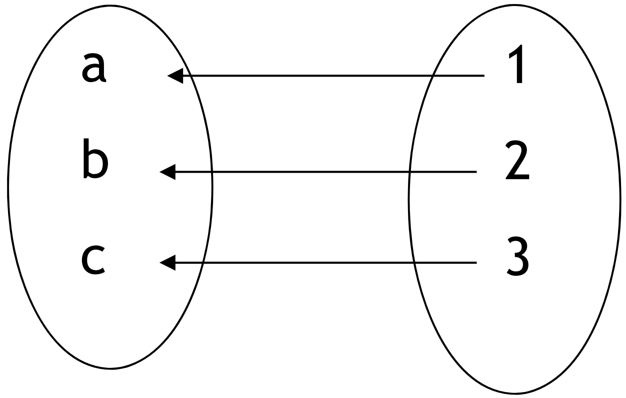
Draw an arrow diagram for f^{-1}.

X = Domain of *f* $\xrightarrow{f}$ **Y = Co-domain of *f***



X = Domain of *f* $\xleftarrow{f^{-1}}$ **Y = Co-domain of *f***

X = Co-domain of f^{-1} **Y = domain of f^{-1}**



Example (Finding Inverse Function)

Given that $f(x) = x + 3$ is one-to-one and onto,
find the formula for its inverse function.

Solution:

Step 1: Let y = f(x).	$y = x + 3$
Step 2: Make x the subject.	$x = y - 3$
Step 3: Since $y = f(x)$ **=>** $x = f^{-1}(y)$	$f^{-1}(y) = y - 3$
Step 4: Replace y by x	$f^{-1}(x) = x - 3$

Exercise (Finding Inverse Function)

Given that $f(x) = 6x + 5$ is one-to-one and onto, find the formula for its inverse function.

Solution:

Step 1: Let y = f(x)

$$y = 6x + 5$$

Step 2: Make x the subject.

$$6x = y - 5$$

$$x = \frac{y-5}{6}$$

Step 3: Since y = f(x) **=>** x = $f^{-1}(y)$

$$f^{-1}(y) = \frac{y-5}{6}$$

Step 4: Replace y by x

$$f^{-1}(x) = \frac{x-5}{6}$$

Part 3 Common Functions in Computing

Section 1
Introduction

- Introduction to Common Mathematical Functions
- Introduction to Exponential & Logarithmic Functions

Introduction

This section presents various mathematical functions which appear often in the analysis of algorithms, and in computer science.

- Floor & Ceiling Functions;
- Integer Function;
- Absolute Value Function;
- Modular Function (Remainder Function).

We also discuss the exponential and logarithmic functions and their relationship.

Section 2
Common Mathematical Functions

- Floor and Ceiling Functions
- Integer Function
- Absolute Value Functions
- Modular Function (Remainder Function)

Floor and Ceiling Functions

Let *x* be any real number.
Then *x* lies between two integers called the floor and the ceiling of *x*.

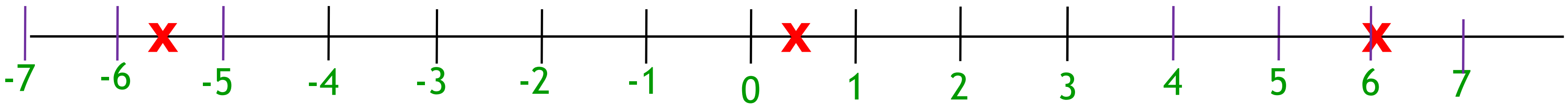
$\lfloor x \rfloor$ the floor of *x*
- Denotes the greatest integer that does not exceed *x*

$\lceil x \rceil$ the ceiling of *x*
- Denotes the least integer that is not less than *x*

Note that
If x is an integer, then $\lfloor x \rfloor = \lceil x \rceil$

Otherwise $\lfloor x \rfloor + 1 = \lceil x \rceil$

Example (Floor and Ceiling Functions)



$\lfloor -5.5 \rfloor$ = -6

$\lceil -5.5 \rceil$ = -5

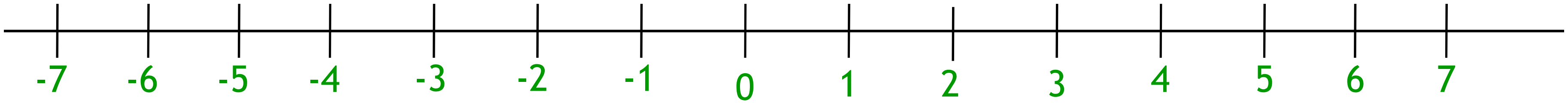
$\lfloor 0.5 \rfloor$ = 0

$\lceil 0.5 \rceil$ = 1

$\lfloor 6 \rfloor$ = 6

$\lceil 6 \rceil$ = 6

Exercise (Floor & Ceiling Functions)



$\lfloor -6.5 \rfloor =$ - 7

$\lceil -6.5 \rceil =$ - 6

$\lfloor \sqrt{2} \rfloor =$ 1

$\lceil \sqrt{2} \rceil =$ 2

$\lfloor 5.12 \rfloor =$ 5

$\lceil 5.12 \rceil =$ 6

$\lfloor 7 \rfloor =$ 7

$\lceil 7 \rceil =$ 7

Integer Function

Let *x* be any real number.
The integer value of *x*, *INT(x)* ,
converts *x* into an integer
by <u>deleting</u> the fractional part of the number.

Example (Integer Function)

INT(395.01) = 395

INT(395.99) = 395

INT(- 395) = -395

Exercise (Integer Function)

INT(5.38) = 5

INT(- 6.24) = - 6

INT(9) = 9

Absolute Value Function

The **absolute value** of the real number *x*, is written as **ABS(*x*)** or **|*x*|.**

$|x| = x$ when *x* is a positive number

$|x| = -x$ when *x* is a negative number

Take Magnitude
Ignore + / -

Example (Absolute Value Function)

$| \; 29.33| = \; 29.33$

$| -29.33 | = 29.33$

Exercise (Absolute Value Function)

$|\ 3.25| =$ *3.25*

$|\ -6\ | =$ *6*

$|\ 5\ | =$ *5*

Modular Function (Remainder Function)

The **modular** function returns the **remainder** of **a** $\div$ **b**
and is written as **"mod(*a*, *b*)"** or **"*a* mod *b*"**

Example (Mod Function)

mod (4,3)	=	1
mod (22,3)	=	1
mod (6,7)	=	6
30 mod 2	=	0
0 mod 12	=	0

$$\begin{array}{r|l} & 1 \\ \hline 3 & 4 \\ - & 3 \\ \hline & \boxed{1} \\ \hline \end{array} \qquad \begin{array}{r|l} & 0 \\ \hline 7 & 6 \\ - & 0 \\ \hline & \boxed{6} \\ \hline \end{array} \qquad \begin{array}{r|l} & 15 \\ \hline 2 & 30 \\ - & 30 \\ \hline & \boxed{0} \\ \hline \end{array} \qquad \begin{array}{r|l} & 0 \\ \hline 12 & 0 \\ - & 0 \\ \hline & \boxed{0} \\ \hline \end{array}$$

Exercise (Mod Function)

34 mod 7 = 6

126 mod 4 = 2

4 mod 7 = 4

7 mod 4 = 3

mod (20, 6) + mod (4, 5)

= 4 + 2

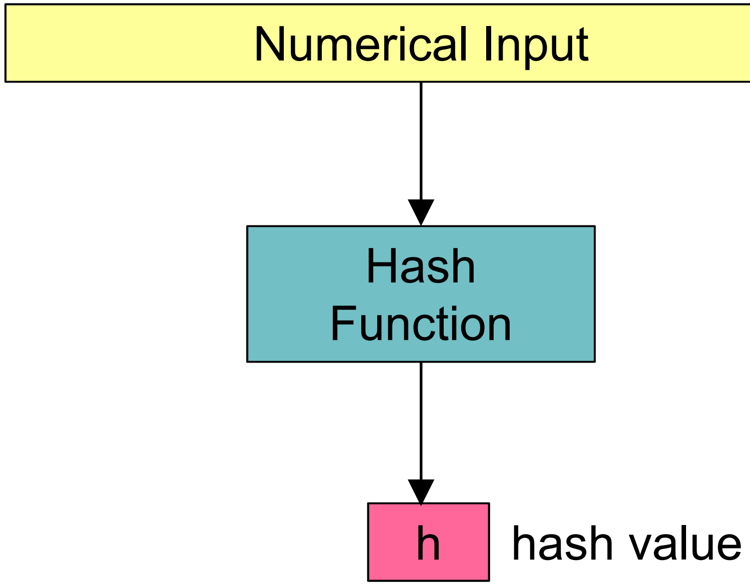
= 6

$$\begin{array}{r} 3 \\ 6\,\overline{)\,20} \\ -\;18 \\ \hline \boxed{2} \end{array} \qquad \begin{array}{r} 0 \\ 5\,\overline{)\,4} \\ -\;0 \\ \hline \boxed{4} \end{array}$$

Application : Hash Function

Hash Function is :

- A function which converts a numerical input into a compressed numerical value (called the hash value),
- frequently using the *mod* function.



Application : Hash Function in Error Checking

Check Digit :

- A check digit is used to detect error in data entry.
- It consists of one or two digits, and is computed by a hash function using the other digits in the sequence input.
- Frequently, the *mod* function is used in a hash function.
- It is used in identification numbers such as :
  - Bank account numbers
  - Credit card numbers
  - Identity card (IC) numbers
  - Car plate numbers
  - International Standard Book Numbers

Application : Hash Function in Error Checking (ISBN - International Standard Book Number)

- ISBN-13 is a unique number assigned to each book.
- The number has 12 information digits and ends with 1 check digit.
  - Format : $\mathbf{d_1\ d_2\ d_3\ d_4\ d_5\ d_6\ d_7\ d_8\ d_9\ d_{10}\ d_{11}\ d_{12} - d_{13}}$
- The check digit, $\mathbf{d_{13}}$, is computed by the following hash function :

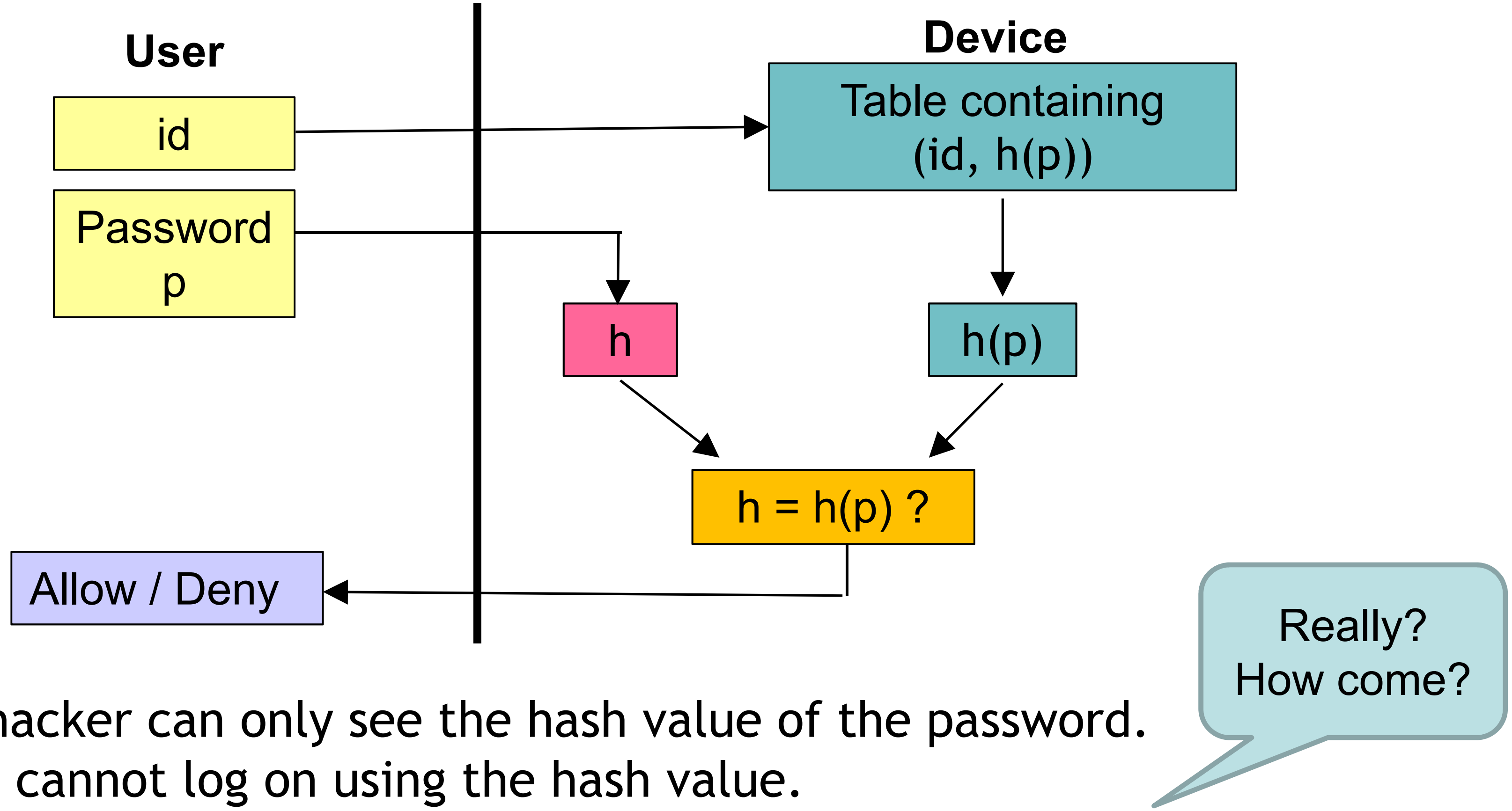
$$\begin{aligned} \text{remainder} &= (\text{ odd-positioned digits} + \text{even-positioned digit} * 3\ ) \bmod 10 \\ &= (\ [d_1+d_3+d_5+d_7+d_9+d_{11}] + [d_2+d_4+d_6+d_8+d_{10}+d_{12}] * 3\ ) \bmod 10 \\ d_{13} &= (10 - \text{remainder}) \bmod 10 \end{aligned}$$

<u>Example : Check Digit of ISBN 978-0-534-49096-6</u>

$$\begin{aligned} \text{remainder} &= (\ [9+8+5+4+9+9] + [7+0+3+4+0+6]*3\ ) \bmod 10 \\ &= 104 \bmod 10 = 4 \\ d_{13} &= (10 - 4) \bmod 10 = 6 \\ &= \text{Last digit of the ISBN} \end{aligned}$$

Application : Hash Function in Cryptography

Hash Function protects password storage.
Instead of storing password as it is, most password file store the hash value of the password. So, it is stored as *ordered pairs* (id, h(p)).



- A hacker can only see the hash value of the password.
- He cannot log on using the hash value.
- He also cannot derive the password based on the hash value.

Section 3
Exponential Function

- Introduction and Graph of Exponential Function
- Laws of Exponents
- Evaluating Using Change of Base
- Evaluating Using Grouping of Base

Exponential Function

An **exponential function** is represented by

$$y = \mathbf{f(x) = b^x},$$

where b is a fixed positive (real) number.

b is called the **base**
x is called the **exponent (or power).**

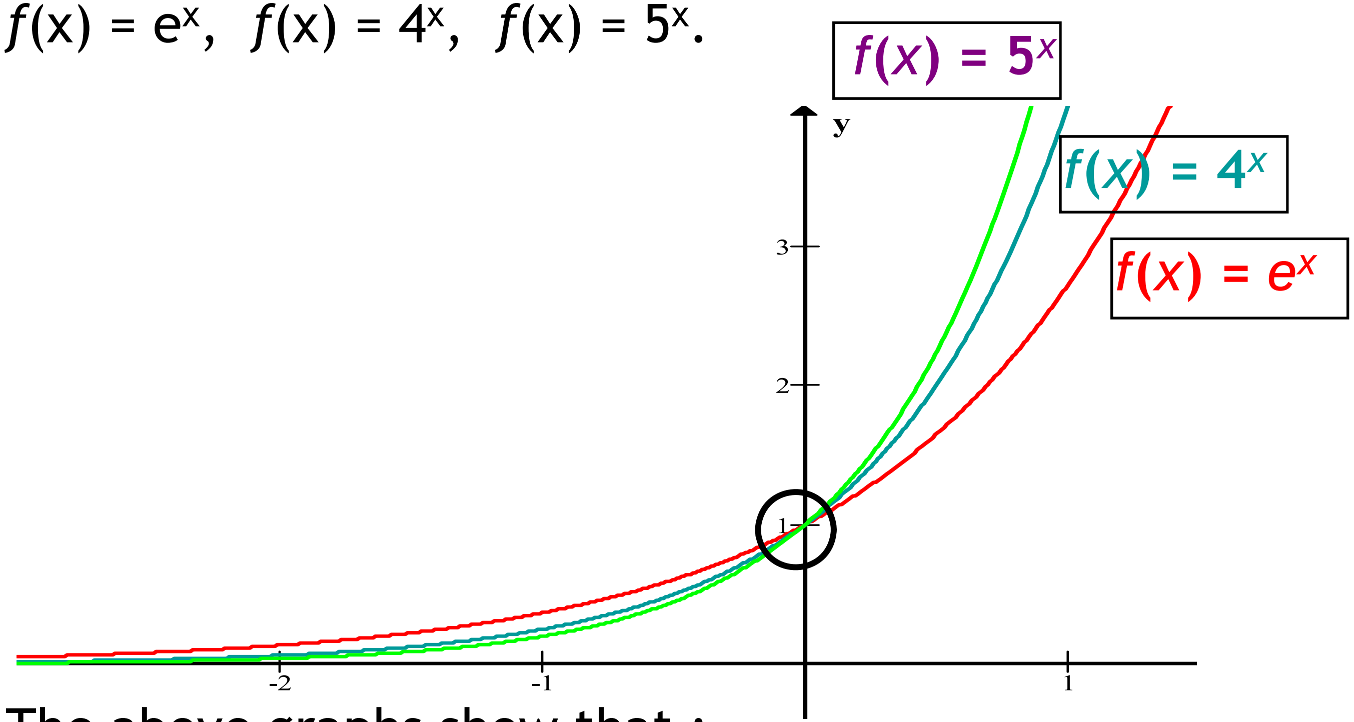
where $\mathbf{b^0 = 1}$ and $\mathbf{b^{-x} = 1/b^x}$.

Examples of exponential functions are:
$y = 10^x$, $y = 4^x$, $y = e^x$, $y = 32^x$

Natural Constant = 2.718

Graph of Exponential Function

Draw the graphs of the exponential functions
$f(x) = e^x$, $f(x) = 4^x$, $f(x) = 5^x$.



The above graphs show that :

1. The <u>curves</u> of all exponential functions have **same shape.**
2. They all **pass through** the point **(0,1)** and
3. Lie completely above the ***x*-axis** (i.e. **asymptote**).

Asymptote = the line that the curves tend to <u>but do not touch</u>

Laws of Exponents

$$a^n \times a^m = a^{n+m}$$

$$\left(a^n\right)^m = a^{nm}$$

$$\frac{a^n}{a^m} = a^{n-m}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$a^{-n} = \frac{1}{a^n}$$

Example (Laws of Exponents)

(a)	$2010^0 = 1$	$b^0 = 1$
(d)	$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$	$a^{-n} = \frac{1}{a^n}$
(g)	$7^2 \times 7^3 = 7^{2+3} = 7^5$	$a^n \times a^m = a^{n+m}$
(i)	$(2^2)^3 = 2^{2 \times 3} = 2^6$	$(a^n)^m = a^{n\,m}$
(k)	$(3x)^3 = 3^3 x^3 = 27x^3$	$(ab)^n = a^n b^n$
(m)	$\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3} = \frac{8}{27}$	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
(o)	$\frac{5^6}{5^4} = 5^{6-4} = 5^2 = 25$	$\frac{a^n}{a^m} = a^{n-m}$

Exercise (Laws of Exponents)

(a) $2010^0 = 1$

(b) $23x^0 = 23(1) = 23$

(c) $(31)^0 = 1$

(d) $5^{-2} = \dfrac{1}{5^2} = \dfrac{1}{25}$

(e) $x^{-3} = \dfrac{1}{x^3}$

(f) $(4)^{-2} = \dfrac{1}{4^2} = \dfrac{1}{16}$

Exercise (Laws of Exponents) (Cont'd)

(g) $7^2 \times 7^3 = 7^5$

(h) $x^{3/2} \times x^{5/2} = x^{8/2} = x^4$

(i) $(2^2)^3 = 2^6$

(j) $(x^{1/3})^6 = x^2$

(k) $(3x)^3 = 3^3 x^3 = 27x^3$

(l) $(3 \times 2)^3 = 3^3 \times 2^3$

$=(27)(8) = 216$

Exercise (Laws of Exponents) (Cont'd)

(m) $\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3} = \frac{8}{27}$

(n) $\left(\frac{4}{3}\right)^3 = \frac{4^3}{3^3} = \frac{64}{27}$

(o) $\frac{5^6}{5^4} = 5^{6-4} = 5^2 = 25$

(p) $\frac{3x}{\sqrt{3x}} = \frac{3x}{(3x)^{\frac{1}{2}}} = (3x)^{\frac{1}{2}} = \sqrt{3x}$

Example (Change of Base)

Evaluate the following:

(i) $4^2 + 64^{2/3}$

$= 16 + \left(4^3\right)^{\frac{2}{3}}$

$= 16 + (4)^2$

$= 32$

(ii) $7^3 \times 7^{-4} \times 7^5$

$= 7^{3-4+5}$

$= 7^4$

$= 2401$

(iii) $(⅓)^{-2} - 5^0$

$= (3^{-1})^{-2} - 1$

$= 3^2 - 1$

$= 8$

Exercise (Change of Base)

Simplify the expression $81^2 \times \frac{1}{9} \times 27^{-1} \times 3^2$

in exponential form with base 3.

Solution :

$$81^2 \times \frac{1}{9} \times 27^{-1} \times 3^2$$

$$= (3^4)^2 \times \frac{1}{3^2} \times (3^3)^{-1} \times 3^2$$

$$= (3^4)^2 \times 3^{-2} \times (3^3)^{-1} \times 3^2$$

$$= 3^8 \times 3^{-2} \times 3^{-3} \times 3^2$$

$$= 3^{8-2-3+2} = 3^5 = 243$$

Example (Grouping of Base)

Simplify the following expressions, leaving your answers in only positive indices:

$(q^5 r^3) (p^4 q^{-2} r^2)$

$= \quad p^4 \quad q^{5-2} \quad r^{3+2}$

$= \quad p^4 \quad q^3 \quad r^5$

Exercise (Grouping of Base)

Simplify the following expressions, leaving your answers in only positive indices:

$$\frac{x^4 y^2 z^{-2}}{y^{-2}(xz)^3}$$

$$= \frac{x^4 y^2 z^{-2}}{y^{-2} x^3 z^3}$$

$$= x^{4-3} \quad y^{2-(-2)} \quad z^{-2-3}$$

$$= x \quad y^4 \quad z^{-5}$$

$$= \frac{xy^4}{z^5}$$

Section 4
Logarithmic Function

- Introduction
- Laws of Logarithmic Functions
- Logarithm of Different Bases
- Conversion to Other Bases
- Relationship Between Logarithmic and Exponential Functions

Introduction

Why the Need for Logarithmic Function?

Some calculations involving multiplication, division & exponentiation – complex

Log Function – *used to* reduce computing time.

Laws of Logarithmic Functions

$$\log_b 1 = 0 \ , \ \ b > 0$$

$$\log_b b = 1 \ , \ \ b \neq 0$$

$$\log_b MN \ = \ \log_b M + \log_b N$$

$$\log_b \left(\frac{M}{N} \right) \ = \ \log_b M - \log_b N$$

$$\log_b N^k = k \log_b N$$

Example (Laws of Logarithmic Functions)

$$\log_b 1 = 0 \ , \ b > 0$$

$\log_2 1 = \log_{3.2} 1 = 0$

$$\log_b b = 1 \ , \ b \neq 0$$

$\log_2 2 = \log_{3.2} 3.2 = 1$

$$\log_b MN = \log_b M + \log_b N$$

$\log_2 (5x3) = \log_2 5 + \log_2 3$

$$\log_b \left(\frac{M}{N} \right) = \log_b M - \log_b N$$

$\log_2 (\frac{5}{3}) = \log_2 5 - \log_2 3$

$$\log_b N^k = k \log_b N$$

$\log_2 9^4 = 4 \log_2 9$

Exercise (Laws of Logarithmic Functions)

(a) $\log_4 1 = 0$

(b) $20 (\log_{25} 25) = 20(1) = 20$

(c) $\log_{25} 25^{-3} = -3$

(d) $6\log_{10} 100 = 6\log_{10} 10^2 = 6(2) = 12$

(e) $\log_3 \left(\dfrac{4}{7}\right) = \log_3 4 - \log_3 7$

(f) $12(\log_2 32^3) = 12\,(\mathbf{3}\log_2 32)$

$= 36\,(\log_2 32)$

$= 36\,(\log_2 2^5)$

$= 36\,(5)(1) = 180$

Example (Evaluating Logarithms)

Use the laws of logarithm to simplify the following expressions:

$$3\log_3 3 \quad + \quad \log_4 1 \quad - \quad \log_6 6^8$$

$\boxed{\log_b b = 1}$ $\boxed{\log_b 1 = 0}$ $\boxed{\log_b N^k = k\log_b N}$

$$= \; 3 \quad (1) \quad + \quad 0 \quad - \; 8\log_6 6$$

$$= \; 3 - 8\,(1)$$

$\boxed{\log_b b = 1}$

$$= \; -5$$

Exercise (Evaluating Logarithms)

Use the laws of logarithm to simplify the following expressions:

$$\log_2 (x+1) - 3\log_2(x^2) + \log_2 (x-1)^4$$

$$= \log_2 (x+1) - \log_2 (x^2)^3 + \log_2 (x-1)^4$$

$$= \log_2 \frac{(x+1)(x-1)^4}{(x^2)^3}$$

$$= \log_2 \frac{(x+1)(x-1)^4}{x^6}$$

Logarithms of Different Bases

Logarithms can be expressed to **any positive** base,

but bases 10 and *e* are used for easier calculation
because scientific **calculators** have these two log functions.

Hence, log to **other bases** are **changed** to **base 10 or** *e* to facilitate further calculations.

Log to base 10 = $\log_{10}$ = **log** = **lg** = *common logarithm*
Log to base *e* = $\log_e$ = **ln** = *natural logarithm*

<u>Note</u>
e = *natural constant* = 2.718281828...

ln e = $\log_e e$ = 1
log 10 = $\log_{10} 10$ = 1

Conversion to Other Bases

To convert logarithm with other bases to base 10 or *e*,

use the following formula:

$$\log_b M = \frac{\lg M}{\lg b} = \frac{\ln M}{\ln b}$$

Example (Conversion to Other of Bases)

Calculate $\log_7 16.8$.

Solution:

Converting to base 10

$$\log_7 16.8 = \frac{\lg 16.8}{\lg 7} = 1.45$$

Converting to base e

$$\log_7 16.8 = \frac{\ln 16.8}{\ln 7} = 1.45$$

Exercise (Conversion to Other Bases)

Find the value of the following:

(i) $\log_3 53$

$= \dfrac{\ln 53}{\ln 3} \; or \; \dfrac{\lg 53}{\lg 3}$

$= 3.61$

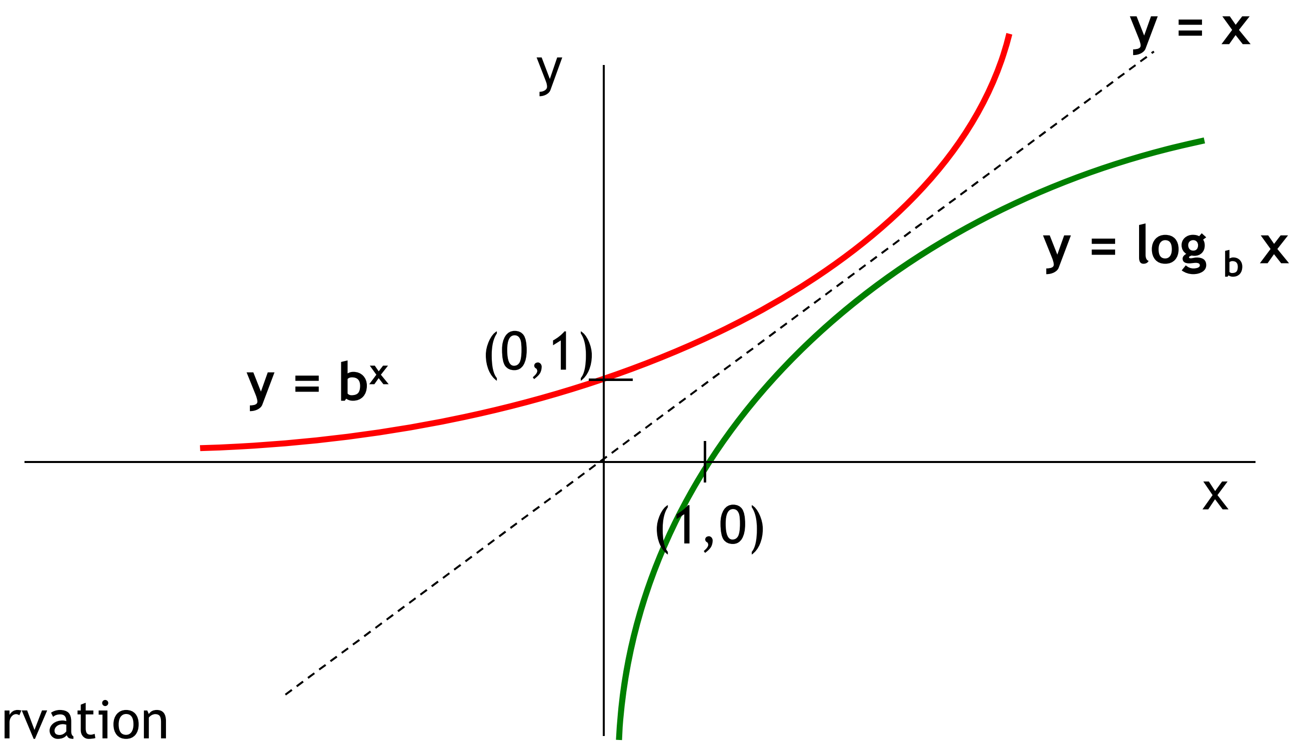
(ii) $\log_{2.5} 3.9$

$= \dfrac{\ln 3.9}{\ln 2.5} \; or \; \dfrac{\lg 3.9}{\lg 2.5}$

$= 1.49$

Relationship between Exponential & Logarithmic Functions

They are inverse of each other (mirror image)



Observation

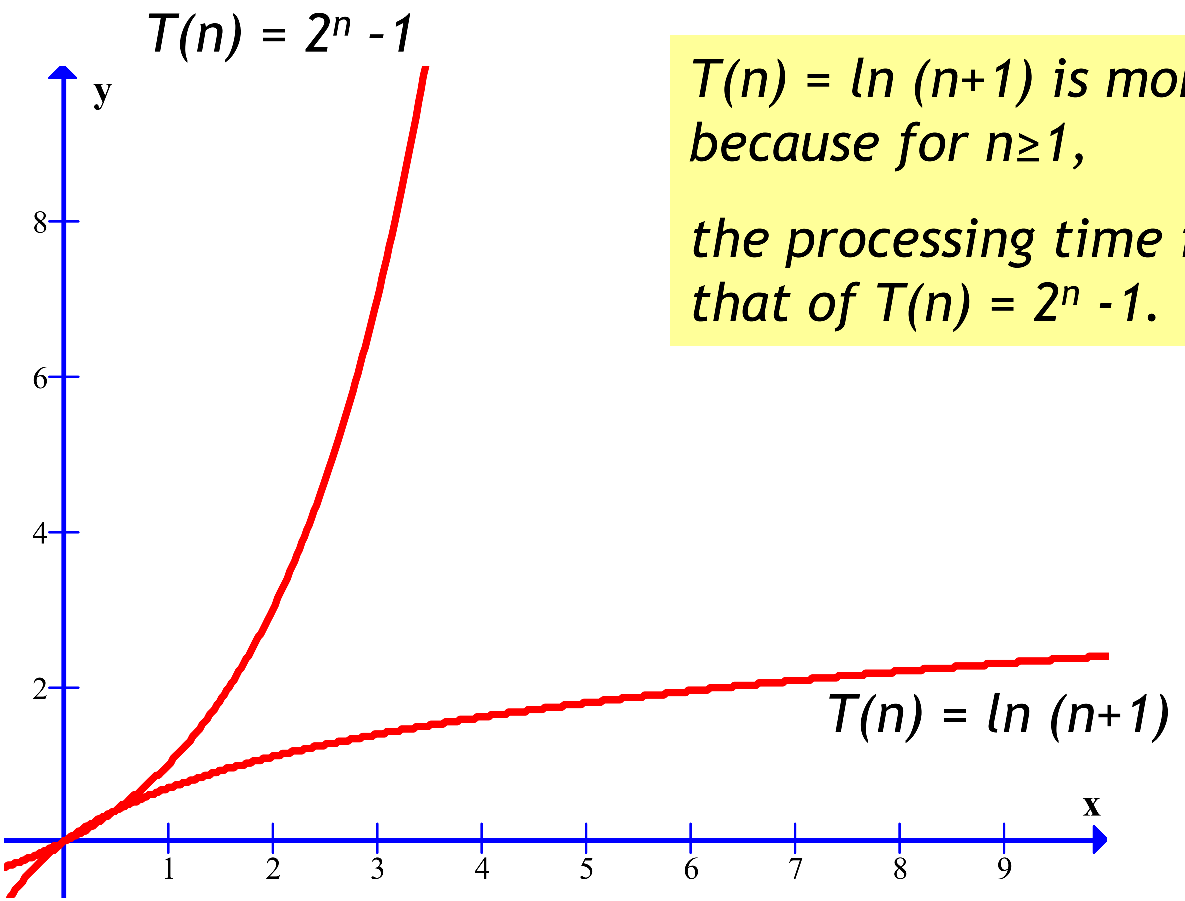
1. Exponential function $y = b^x$ is always positive.
2. What is the domain of log? **x > 0**

   Log of -ve no. = ? **(Not defined)**

   Log 0 = ? **(Not defined)**

Application : Algorithm Efficiency

Given that n represents the number of inputs and T(n) represents the time taken to process n inputs, which of the given functions below would be the more efficient in terms of processing time?



$T(n) = \ln(n+1)$ is more efficient because for $n \geq 1$,

the processing time is lower than that of $T(n) = 2^n - 1$.

END OF TOPIC